

Exercise J02

Q1. Prove that $\cot A + \cot B + \cot C = \frac{a^2 + b^2 + c^2}{4\Delta}$

Ans. -

Sol.
$$\begin{aligned}\cot A + \cot B + \cot C &= \frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C} \\ &= \frac{b^2 + c^2 - a^2}{2bc \sin A} + \frac{a^2 + c^2 - b^2}{2ac \sin B} + \frac{a^2 + b^2 - c^2}{2ab \sin C} \\ &= \frac{a^2 + b^2 + c^2}{4\Delta}\end{aligned}$$

Q2. Let area of a triangle ABC is $\frac{\sqrt{3}-1}{2}$ where AC = 2, AB = $(\sqrt{3}-1)$ and $\angle A$ is acute. Find the value of $(BC)^2$.

Ans. 2

Sol.
$$\Delta = \frac{1}{2} AC \cdot AB \sin A$$

$$\frac{\sqrt{3}-1}{2} = \frac{1}{2} (2) (\sqrt{3}-1) \sin A$$

$$\sin A = \frac{1}{2}$$

$A = 30^\circ$ (A is acute)

$$\cos A = \frac{AB^2 + AC^2 - BC^2}{2 \cdot AB \cdot AC}$$

$$\frac{\sqrt{3}}{2} = \frac{(\sqrt{3}-1)^2 + (2)^2 - BC^2}{2 \cdot 2(\sqrt{3}-1)}$$

$$2(3 - \sqrt{3}) = 3 + 1 - 2\sqrt{3} + 4 - BC^2$$

$$6 - 2\sqrt{3} = 8 - 2\sqrt{3} - BC^2$$

$$BC^2 = 2$$

Q3. Prove that $a(\cos C - \cos B) = 2(b - c) \cos^2 \frac{A}{2}$

Ans. -

Sol.
$$a(\cos C - \cos B) = 2(b - c) \cos^2 \frac{A}{2}$$

$$\text{LHS.} = 2R \sin A \left(2 \sin \left(\frac{B+C}{2} \right) \cdot \sin \left(\frac{B-C}{2} \right) \right)$$

$$= 2R^2 \sin \frac{A}{2} \cdot 2 \cos \frac{A}{2} \cdot \sin \left(\frac{B-C}{2} \right)$$

$$= 2R \cdot 2 \cos^2 \frac{A}{2} \left[2 \cos \left(\frac{B+C}{2} \right) \cdot \sin \left(\frac{B-C}{2} \right) \right]$$

$$= 2 \cos^2 \frac{A}{2} \cdot 2R [\sin B - \sin C]$$

$$= 2 \cos^2 \frac{A}{2} (b - c) = \text{RHS}$$

Q4. Prove that $C^2 = (a - b)^2 \cos^2 \frac{C}{2} + (a + b)^2 \sin^2 \frac{C}{2}$

Ans. -

Sol. $\text{RHS} = (a - b)^2 \cos^2 \frac{C}{2} + (a + b)^2 \sin^2 \frac{C}{2}$

$$= a^2 \cos^2 \frac{C}{2} + b^2 \cos^2 \frac{C}{2} - 2ab \cos^2 \frac{C}{2} + a^2 \sin^2 \frac{C}{2} + b^2 \sin^2 \frac{C}{2} + 2ab \sin^2 \frac{C}{2}$$

$$= a^2 + b^2 - 2ab \left(\cos^2 \frac{C}{2} - \sin^2 \frac{C}{2} \right)$$

$$= a^2 + b^2 - 2ab \cos C$$

$$= a^2 + b^2 - 2ab \left(\frac{a^2 + b^2 - c^2}{2ab} \right)$$

$$= C^2$$

Q5. Prove that $\frac{a \sin(B - C)}{b^2 - c^2} = \frac{b \sin(C - A)}{c^2 - a^2} = \frac{c \sin(A - B)}{a^2 - b^2}$

Ans. -

Sol. $\frac{a \sin(B - C)}{b^2 - c^2}$

$$= \frac{2R \sin A \cdot \sin(B - C)}{4R^2 (\sin^2 B - \sin^2 C)}$$

$$= \frac{2R \sin A \cdot \sin(B - C)}{4R^2 \sin(B + C) \cdot \sin(B - C)} = \frac{1}{2R}$$

$$= \frac{b \sin(C - A)}{c^2 - a^2}$$

$$= \frac{2R \sin B \cdot \sin(C - A)}{4R^2 [\sin^2(A + C) \cdot \sin(C - A)]} = \frac{1}{2R}$$

Q6. Prove that $a^2 (\cos^2 B - \cos^2 C) + b^2 (\cos^2 C - \cos^2 A) + c^2 (\cos^2 A - \cos^2 B) = 0$

Ans. -

Sol. $a^2 (\cos^2 B - \cos^2 C) = a^2 (\sin^2 C - \sin^2 B)$

$$= a^2 \left(\frac{c^2}{4R^2} - \frac{b^2}{4R^2} \right)$$

$$= \frac{1}{4R^2} [a^2 c^2 - a^2 b^2] \dots\dots\dots(1)$$

Similarly

$$b^2 (\cos^2 C - \cos^2 A) = \frac{1}{4R^2} [b^2 a^2 - b^2 c^2] \dots\dots\dots(2)$$

$$c^2 (\cos^2 A - \cos^2 B) = \frac{1}{4R^2} [c^2 b^2 - c^2 a^2] \dots\dots\dots(2)$$

$$(1) + (2) + (3)$$

$$\text{RHS} = 0$$

Q7. Prove that $\frac{abc}{s} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \Delta$

Ans. -

$$\begin{aligned} \text{Sol. } & \frac{abc}{s} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} \\ &= \frac{abc}{s} \sqrt{\frac{s(s-a)}{bc}} \sqrt{\frac{s(s-b)}{ac}} \sqrt{\frac{s(s-c)}{ab}} \\ &= \sqrt{s(s-a)(s-b)(s-c)} = \Delta \end{aligned}$$

Q8. Prove that $\frac{(a+b+c)^2}{a^2+b^2+c^2} = \frac{\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}}{\cot A + \cot B + \cot C}$

Ans. -

$$\begin{aligned} \text{Sol. } & \frac{\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}}{\cot A + \cot B + \cot C} \\ &= \frac{\frac{s(s-a)}{\Delta} + \frac{s(s-b)}{\Delta} + \frac{s(s-c)}{\Delta}}{\frac{\cos A}{\sin A} + \frac{\cos B}{\sin B} + \frac{\cos C}{\sin C}} \\ &= \frac{\frac{s}{\Delta} (3s - (a+b+c))}{\frac{b^2+c^2-a^2}{2bc\sin A} + \frac{a^2+c^2-b^2}{2ac\sin B} + \frac{a^2+b^2-c^2}{2ab\sin C}} \\ &= \frac{\frac{s^2}{\Delta}}{\frac{a^2+b^2+c^2}{4\Delta}} = \frac{(2s)^2}{a^2+b^2+c^2} \end{aligned}$$

Q9. Check the ambiguity .

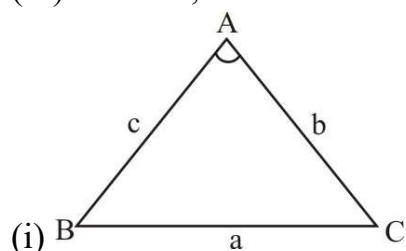
(i) If $a = 5$, $b = 7$ and $\sin A = \frac{3}{4}$

(ii) $2b = 3a$, $\tan^2 A = \frac{3}{5}$

(iii) $A = 30^\circ$, $b = 8$ and $a = 6$

(iv) $B = 30^\circ$, $C = 150^\circ$, $b = 75$

Ans.



$$\cos A = \frac{a^2 + c^2 - b^2}{2bc}$$

$$C^2 - (2b \cos A)C + b^2 - a^2 = 0$$

$$D = 4b^2 + \cos^2 A - 4b^2 + 4a^2$$

$$D = 4(a^2 - b^2 \sin^2 A)$$

$$a = 5, b = 7, \sin A = \frac{3}{4}$$

$$D = 4\left(25 - 49 \cdot \frac{9}{16}\right) = (-) \text{ive}$$

No such triangle is possible

$$(iii) a = 6, b = 8, A = 30^\circ \sin 30 = \frac{1}{2}$$

$$D = 4(a^2 - b^2 \sin^2 A)$$

$$= \left(36 - 64 \cdot \frac{1}{4}\right) = (+) \text{ive}$$

$$b^2 - c^2 = \text{product of } c_1 \text{ \& } c_2 = 64.36 (+) \text{ive}$$

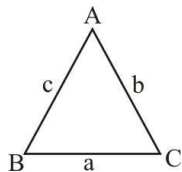
2 triangles are possible

Sol. -

Q10. In the ambiguous case given a, b and A . Prove that the differences between the values of ' c ' is $2\sqrt{a^2 - b^2 \sin^2 A}$.

Ans. -

Sol.



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$C^2 - (2b \cos A)C + b^2 - a^2 = 0$$

$$C_1 \text{ \& } C_2 = \frac{2b \cos A \pm \sqrt{4b^2 \cos^2 A - 4b^2 + 4a^2}}{2}$$

$$= b \cos A \pm \sqrt{a^2 - b^2 \sin^2 A}$$

Q11. Prove that $(r_1 + r_2) \tan \frac{C}{2} = (r_3 - r) \cot \frac{C}{2} = c$

Ans. -

Sol.

$$(i) (r_1 + r_2) \tan \frac{C}{2}$$

$$= \left(\frac{\Delta}{s - c} + \frac{\Delta}{s - b} \right) \cdot \frac{\Delta}{s(s - c)}$$

$$= \frac{\Delta^2(s - b + s - c)}{s(s - b)(s - c)} = \frac{\Delta^2(2s - a - b)}{\Delta^2} = c$$

$$\begin{aligned}
& (r_1+r_2) \cot \frac{C}{2} \\
&= \left(\frac{\Delta}{s-c} - \frac{\Delta}{s} \right) \cdot \frac{s(s-c)}{\Delta} \\
&= \frac{\Delta(s-s+c)}{s(s-c)} = \frac{s(s-c)}{\Delta^2} = C
\end{aligned}$$

$$(ii) \ b = \frac{3}{2} a, \tan^2 A = \frac{3}{5}$$

$$\tan A = \frac{\sqrt{3}}{\sqrt{5}}$$

$$\sin A = \frac{\sqrt{3}}{2\sqrt{2}}$$

$$D = 4(a^2 - b^2 \sin^2 A)$$

$$= 4 \left(a^2 - \frac{9}{4} a^2 \cdot \frac{3}{8} \right)$$

$$= \left(1 - \frac{27}{32} \right) = (+) \text{ive}$$

$$\text{product } C_1 \ \& \ C_2 = b^2 - a^2$$

$$= \frac{9}{4} a^2 - a^2 = \frac{5}{4} a^2 = (+) \text{ive}$$

two triangles are possible

$$(iv) \ c = 150, b = 75, \angle B = 30^\circ$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$a^2 - (2C \cos B) a + c^2 - b^2 = 0$$

$$D = 4c^2 \sin^2 B - 4c^2 + 4b^2$$

$$= 4(b^2 - c^2 \sin^2 B)$$

$$= 4 \left((75)^2 - (150)^2 \cdot \frac{1}{4} \right) = 0$$

One right angle triangle is possible $\angle C = 90$

Q12. Prove that $(r_1 - r)(r_2 - r)(r_3 - r) = 4Rr^2$

Ans. -

Sol. $(r_1 - r)(r_2 - r)(r_3 - r)$

$$= \left(\frac{\Delta}{s-a} - \frac{\Delta}{s} \right) \left(\frac{\Delta}{s-b} - \frac{\Delta}{s} \right) \left(\frac{\Delta}{s-c} - \frac{\Delta}{s} \right)$$

$$= \frac{\Delta a}{s(s-a)} \cdot \frac{\Delta b}{s(s-b)} \cdot \frac{\Delta c}{s(s-c)}$$

$$\begin{aligned}
&= \frac{\Delta a^3}{s^2 s(s-a)(s-b)(s-c)} \\
&= 4 \frac{\Delta^3 abc}{s^2 s(s-a)(s-b)(s-c)} \\
&= \frac{\Delta^2 abc}{s^2 \Delta^2} \\
&= 4 \left(\frac{\Delta}{s} \right)^2 \frac{abc}{4\Delta} \\
&= 4r^2 R
\end{aligned}$$

Q13. Prove that $(r + r_1) \tan \frac{B-C}{2} + (r + r_2) \tan \frac{C-A}{2} + (r + r_3) \tan \frac{A-B}{2} = 0$

Ans. -

Sol.

$$\begin{aligned}
&= (r_1 + r) \tan \left(\frac{B-C}{2} \right) \\
&= \left(\frac{\Delta}{s} + \frac{\Delta}{s-a} \right) \cdot \left(\frac{b-c}{b+c} \right) \cot \frac{A}{2} \\
&= \frac{\Delta(2s-a)}{s(s-a)} \cdot \left(\frac{b-c}{b+c} \right) \cdot \frac{s(s-a)}{\Delta} \\
&= \left(\frac{b+c}{b+c} \right) \cdot (b-c) = b-c \dots\dots\dots(1)
\end{aligned}$$

Similarly $(r + r_2) \tan \left(\frac{C-A}{2} \right) = c-a \dots\dots\dots(2)$

$$(r + r_3) \tan \left(\frac{A-B}{2} \right) = a-b \dots\dots\dots(3)$$

add (1), (2), (3)

$$\text{RHS} = 0$$

Q14. Prove that $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}$

Ans. -

Sol.

$$\begin{aligned}
&\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} \\
&= \frac{s^2}{\Delta^2} + \frac{(s-a)^2}{\Delta^2} + \frac{(s-b)^2}{\Delta^2} + \frac{(s-c)^2}{\Delta^2} \\
&= \frac{4s^2 - 2s(a+b+c) + a^2 + b^2 + c^2}{\Delta^2} \\
&= \frac{4s^2 - 2s(2s) + a^2 + b^2 + c^2}{\Delta^2} \\
&= \frac{a^2 + b^2 + c^2}{\Delta^2}
\end{aligned}$$

Q15. Prove that $(r_3 + r_1)(r_3 + r_2) \sin C = 2 r_3 \sqrt{r_2 r_3 + r_3 r_1 + r_1 r_2}$

Ans. -

Sol. $(r_3 + r_1)(r_3 + r_2) \sin C$

$$\begin{aligned}
 &= \left(\frac{\Delta}{s-c} + \frac{\Delta}{s-b} \right) \left(\frac{\Delta}{s-c} + \frac{\Delta}{s-a} \right) \sin C \\
 &= \Delta \left(\frac{2s-a-c}{(s-a)(s-c)} \right) \Delta \left(\frac{2s-a-c}{(s-a)(s-c)} \right) \sin C \\
 &= \frac{\Delta^2 \cdot ab \cdot \sin C \cdot s}{(s-a)(-a)(s-b)(s-c) \cdot s} \\
 &= \frac{\Delta^2 \cdot 2\Delta \cdot s}{\Delta^2 \cdot (s-c)} \\
 &= 2 \frac{\Delta}{(s-c)} \cdot s \\
 &= 2r_3 \sqrt{r_1 + r_2 + r_3 + r_3 r_3 r_1} \quad [\because \sum r_1 r_2 = s^2]
 \end{aligned}$$

Q16. Prove that $\frac{bc - r_2 r_3}{r_1} = \frac{ca - r_3 r_1}{r_2} = \frac{ab - r_1 r_2}{r_3} = r$

Ans. -

Sol. $\frac{bc - r_2 r_3}{r_1}$

$$\begin{aligned}
 &= \frac{1}{r_1} \left[bc - \frac{\Delta}{(s-c)} \cdot \frac{\Delta}{(s-a)} \right] \\
 &= \frac{1}{r_1} [bc - s(s-a)] \\
 &= \frac{1}{4r_1} [4bc - (2s)(2s-2a)] \\
 &= \frac{1}{4r_1} [4bc - (b+c+a)(b+c-a)] \\
 &= \frac{1}{4r_1} [4bc - (b+c)^2 + a^2] \\
 &= \frac{1}{4r_1} [a^2 - (b-c)^2] \\
 &= \frac{1}{4r_1} [(a+b-c)(a+c-b)] \\
 &= \frac{1}{4r_1} (2s-2c)(2s-2b) \\
 &= \frac{(s-c)(s-b(s-b))}{\Delta} \\
 &= \frac{\Delta}{s} = r \\
 \frac{ca - r_1 r_3}{r_2} &= \frac{1}{r_2} \left[ac - \frac{\Delta}{s-a} \cdot \frac{\Delta}{s-c} \right] \\
 &= \frac{1}{4r_2} [4bc - 2s(2s-2b)]
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{4r_2} [b^2 - (a - c)^2] \\
&= \frac{1}{4r_2} (2s - 2c)(2s - 2a) \\
&= \frac{s - a(s - c)(s - b)}{\Delta} \\
&= \frac{\Delta}{s} r
\end{aligned}$$

Q17. Prove that $Rr (\sin A + \sin B + \sin C) = \Delta$

Ans. -

Sol. $R_r (\sin A + \sin B + \sin C)$

$$\begin{aligned}
&= R_r \left(\frac{a}{2R} + \frac{b}{2r} + \frac{c}{2r} \right) \\
&= R_r \left(\frac{2s}{2R} \right) \\
&= r_s = \Delta
\end{aligned}$$

Q18. Prove that $2R \cos A = 2R + r - r_1$

Ans. -

Sol. $2R + r - r_1$

$$\begin{aligned}
&= 2R + 4R \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} - 4R \sin \frac{A}{2} \cdot \cos \frac{B}{2} \cdot \cos \frac{C}{2} \\
&= 2R + 4R \sin \frac{A}{2} \cdot \left(\sin \frac{B}{2} \cdot \sin \frac{C}{2} - \cos \frac{B}{2} \cdot \cos \frac{C}{2} \right) \\
&= 2R + 4R \sin \frac{A}{2} \cdot \left(-\cos \left(\frac{B+C}{2} \right) \right) \\
&= 2R + \left(1 - 2\sin^2 \frac{A}{2} \right) \\
&= 2R \cos A
\end{aligned}$$

Q19. If $r_1 = r + r_2 + r_3$ then prove that the triangle is a right angled triangle.

Ans. -

Sol. $r_1 = r + r_2 + r_3$

$$\begin{aligned}
r_1 - r &= r_2 + r_3 \\
\frac{\Delta}{s-a} - \frac{\Delta}{s} &= \frac{\Delta}{s-b} + \frac{\Delta}{s-c} \\
\frac{\Delta(s-s+a)}{s(s-a)} &= \frac{\Delta(2s-b-c)}{(s-b)(s-c)} \\
s^2 - sa &= s^2 - s(b+c) + bc
\end{aligned}$$

$$2s(b + c - a) = 2bc$$

$$(a + b + c)(b + c - a) = 2bc$$

$$(c + b)^2 - a^2 = 2bc$$

$$b^2 + c^2 = a^2 \quad \angle A = 90^\circ$$

Q20. If two times the square of the diameter of the circumcircle of a triangle is equal to the sum of the squares of its sides then prove that the triangle is right angled.

Ans. -

Sol. $2(2R)^2 = a^2 + b^2 + c^2$

$$2(4R^2) = 4R^2 [\sin^2 A + \sin^2 B + \sin^2 C]$$

$$= 2 = \left[\frac{1 - \cos 2A}{2} + \frac{1 - \cos 2B}{2} + \frac{1 - \cos 2C}{2} \right]$$

$$\cos 2A + \cos 2B + \cos 2C = -1$$

$$-1 - 4 \cos A \cos B \cos C = -1$$

$$\cos A \cos B \cos C = 0$$

then at least one of them $(\cos A, \cos B, \cos C) = 0$

let $\cos A = 0$ then $\angle A = 90^\circ$

Q21. In acute angled triangle ABC, a semicircle with radius r_a is constructed with its base on BC and tangent to the other two sides. r_b and r_c are defined similarly. If r is the radius of the incircle of triangle ABC then prove that, $\frac{2}{r} = \frac{2}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$.

Ans. -

$$\Delta = \frac{Cr_a}{2} + \frac{br_a}{2}$$

$$\frac{1}{r_a} = \frac{b+c}{2\Delta}$$

$$\begin{aligned} \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c} &= \frac{b+c}{2\Delta} + \frac{a+c}{2\Delta} + \frac{a+b}{2\Delta} \\ &= \frac{2(a+b+c)}{2\Delta} = \frac{2s}{\Delta} = \frac{2}{r} \end{aligned}$$

p_1, p_2, p_3 then prove that $\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$.

Sol.

$$\text{In } \triangle ABD \sin B = \frac{P_1}{C}$$

$$P_1 = \frac{bc}{2R}$$

$$\frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} = \frac{2R}{bc} + \frac{2R}{ac} + \frac{2R}{2b}$$

$$= \frac{2R[a + b + c]}{abc}$$

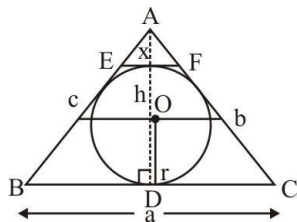
$$= \frac{2R(2s)}{abc} = \frac{s}{\Delta} = \frac{1}{r}$$

$$\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{s-a}{\Delta} + \frac{s-b}{\Delta} + \frac{s-c}{\Delta} = \frac{3s - (a+b+c)}{\Delta}$$

$$= \frac{s}{\Delta} = \frac{1}{r}$$

Ans. -

Sol.



$$EF \parallel BC$$

$$OD = r$$

$$EF = x$$

$$\text{Altitude from vertex 'A'} = h$$

$$\triangle ABC \approx \triangle AEF$$

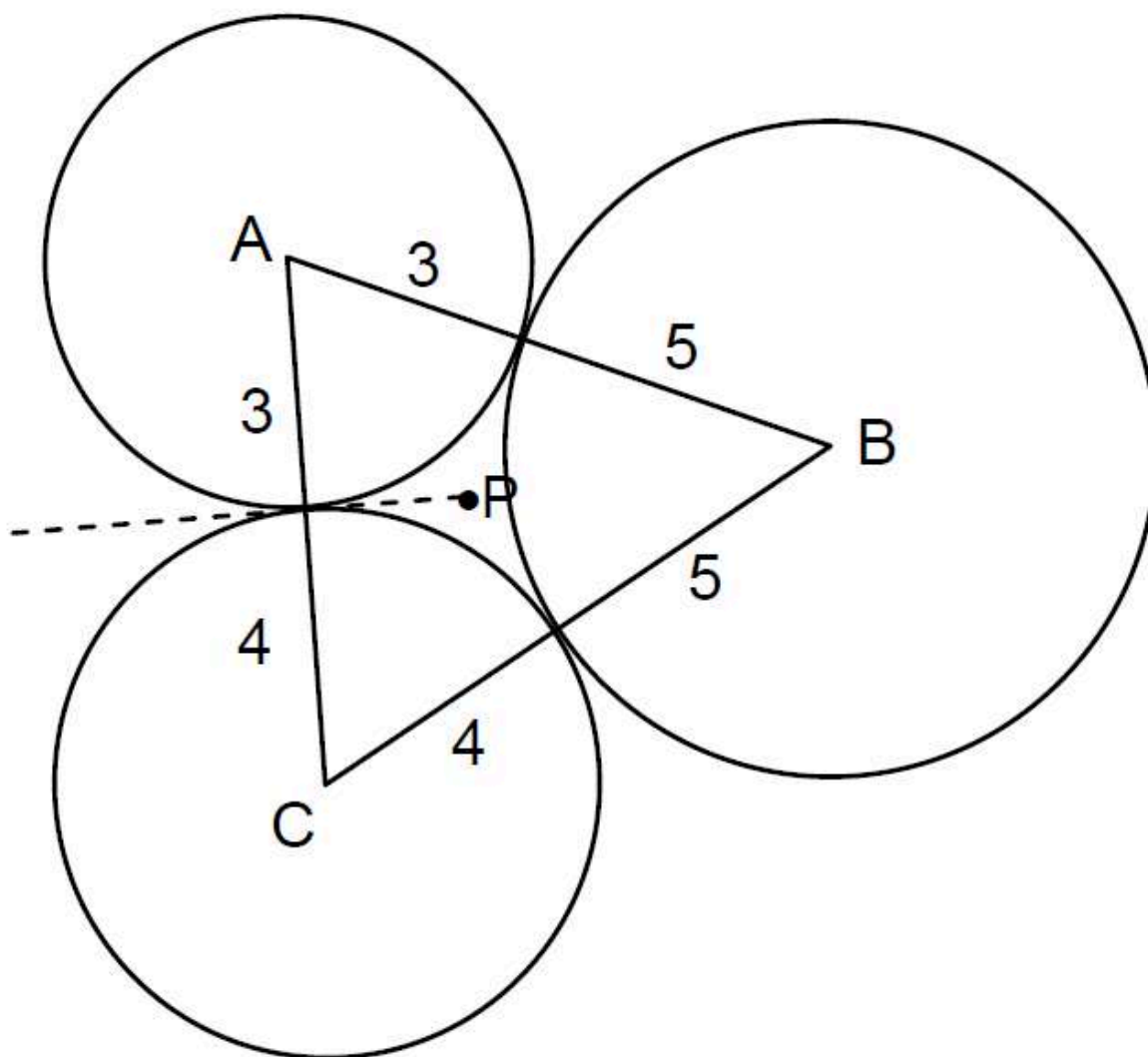
$$\frac{x}{a} = 1 - \frac{a}{s}$$

$$\begin{aligned} \frac{x}{a} + \frac{4}{b} + \frac{7}{c} &= 3 - \frac{(a+b+c)}{s} \\ &= 3 - \frac{2s}{2} = 1 \end{aligned}$$

- Q24. Circles with radii 3, 4 and 5 touch each other externally if P is the point of intersection of tangents to these circles at their points of contact. Find the distance of P from the points of contact.

Ans. -

Sol.



P is incentre of $\triangle ABC$

Distance of P from P.O.C is 'r'

$$S = \frac{7 + 8 + 9}{2} = 12$$

$$r = \frac{\Delta}{s} = \frac{\sqrt{12(5)(4)(3)}}{12} = \sqrt{5}$$

Q25. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If a, b, c (in order) are in arithmetic progression such that $x^2 + 3x + 5 = 0$ and $3x^2 + ax + c = 0$ have a common root, then find the radius of the incircle of triangle ABC.

Ans. 3

Sol. $x^2 + 3x + 5 = 0$ & $3x^2 + ax + c = 0$ (2)

$D = 9 - 20 = (-)$ ive(1)

$\therefore$ Roots of 1st equation are real. So both of equation (1) & (2) are common

$$\frac{1}{3} = \frac{a}{3} = \frac{c}{5}$$

$$a = 9, c = 15$$

$\therefore a, b, c$ in AP

$$2b = a + 5 = 24$$

$$b = 12 \quad s = \frac{9 + 12 + 15}{2} = 18$$

$$r = \frac{\Delta}{s} = \frac{\sqrt{18 \cdot 9 \cdot 6 \cdot 3}}{18} = 3$$

Q26. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. Suppose $a = 6$, $b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

Ans. 3

Sol. $\Delta 15 \sqrt{3}$

$$\angle ACB > 90^\circ$$

$$r^2 = ?$$

$$\Delta = \frac{1}{2} 10 \cdot 6 \cdot \sin C = 15 \sqrt{3}$$

$$\sin C = \frac{\sqrt{3}}{2}$$

$$C = 120$$

$$\cos C = \frac{100 + 36 + c^2}{2 \cdot 10 \cdot 6}$$

$$c = 14$$

$$r = \frac{\Delta}{s}$$

$$s = \frac{a + b + c}{2} = \frac{10 + 6 + 14}{2} = 15$$

$$\therefore r = \sqrt{2} \quad r^2 = 3$$

Q27. In a triangle ABC, let $2a^2 + 4b^2 + c^2 = 2a(2b + c)$, then which of the following holds good?

[Note : All symbols used have usual meaning in a triangle.]

$$(A) \cos B = \frac{-7}{8} \quad (B) \sin(A - C) = 0 \quad (C) \frac{r}{r_1} = \frac{1}{5}$$

$$(D) \sin A : \sin B : \sin C = 1 : 2 : 1$$

Ans. B, C

Sol. $a^2 + 4b^2 - 4b + a^2 + c^2 - 2ac = 0$

$$(a - 2b)^2 + (a - c)(a - 2b)^2 + (a - c)^2 = 0$$

$$a = 2b, \quad a = c$$

$$b = \frac{a}{2}, \quad c = a$$

$$s = \frac{a + b + c}{2} = \frac{2a + \frac{a}{2}}{2} \quad s = \frac{5a}{4}$$

$$(A) \cos B = \frac{a^2 + c^2 - b^2}{2ac} = \frac{a^2 + a^2 - \frac{a^2}{4}}{2a^2} = \frac{(2 - \frac{1}{4})}{2} = \frac{7}{8}$$

$$(B) \therefore a = c \text{ then } \angle A = \angle C$$

$$\sin(A - C) = 0$$

$$(C) \frac{r}{r^1} = \frac{\frac{\Delta}{s}}{\frac{\Delta}{s-a}} = \frac{s-a}{s} = 1 - \frac{a}{s} = 1 - \frac{4}{5} = \frac{1}{5}$$

$$(D) \sin A : \sin B : \sin C = a : b : c$$

$$= 1 : \frac{1}{2} : 1$$

$$= 2 : 1 : 2$$

Q28. In $\triangle ABC$, angles A, B and C are in the ratio $1 : 2 : 3$, then which of the following is(are) correct?

(All symbol used have usual meaning in a triangle.)

$$(A) \text{ Circumradius of } \triangle ABC = c \quad (B) a : b : c = 1 : \sqrt{3} : 2$$

$$(C) \text{ Perimeter of } \triangle ABC = 3 + \sqrt{3} \quad (D) \text{ Area of } \triangle ABC = \frac{\sqrt{3}}{8} c^2$$

Ans. D, B

Sol. $\angle A : \angle B : \angle C = 1 : 2 : 3$

$$\angle A = \frac{\pi}{6} \quad \angle B = \frac{\pi}{3}, \quad \angle C = \frac{\pi}{2}$$

$$\frac{a}{\sin \frac{\pi}{6}} = \frac{b}{\sin \frac{\pi}{3}} = \frac{c}{\sin \frac{\pi}{2}} = 2R$$

$$(B) a : b : c = \frac{1}{2} : \frac{\sqrt{3}}{2} : 1$$

$$(D) \text{ area} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} ab \sin 90$$

$$= \frac{1}{2} \left(\frac{\sqrt{3}}{4} C^2 \right) \dots\dots(1)$$

$$= \frac{\sqrt{3}C^2}{8}$$

(C) Perimeter = can't determine

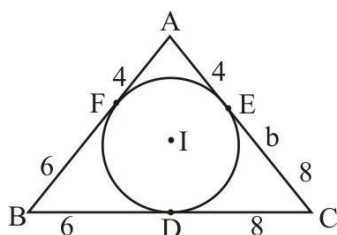
Q29. If the length of tangents from A, B, C to the incircle of $\triangle ABC$ are 4, 6, 8 then which of the following is(are) correct? (All symbols used have usual meaning in a triangle.)

(A) Area of $\triangle ABC$ is $12\sqrt{6}$ (B) r_1, r_2, r_3 are in H.P. (C) a, b, c are in A.P.

(D) $r = \frac{4\sqrt{6}}{3}$

Ans. D, B

Sol.



$$a = 14$$

$$b = 12$$

$$c = 10$$

$$s = \frac{10 + 12 + 14}{2} = 18$$

$$(B) r_1 = \frac{\Delta}{s - a} = \frac{24\sqrt{6}}{4} = 6\sqrt{6}$$

$$r_2 = \frac{\Delta}{s - b} = \frac{24\sqrt{6}}{6} = 4\sqrt{6}$$

$$r_3 = \frac{\Delta}{s - c} = \frac{24\sqrt{6}}{8} = 3\sqrt{6}$$

$$\frac{1}{r_1}, \frac{1}{r_2}, \frac{1}{r_3} = \frac{1}{6\sqrt{6}}, \frac{1}{4\sqrt{6}}, \frac{1}{3\sqrt{6}} \text{ (AP)}$$

(C) a, b, c = (14, 12, 10) in AP

$$(C) r = \frac{24\sqrt{6}}{18} = \frac{4\sqrt{6}}{3}$$

Q30. A triangle ABC has the feature, $\frac{a \cos A + b \cos B + c \cos C}{a + b + c} = \frac{1}{2}$ then the correct statement(s) is/are :

(A) triangle is right angled (B) $R = 2r$ (C) $a = b = c$ (D) $\frac{\sum \sin A}{\prod \sin A} = 2$

Ans. B, C

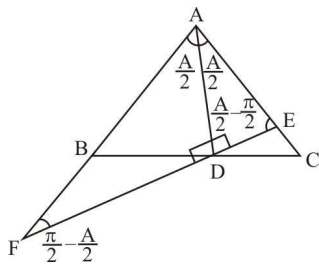
Sol. $2[a \cos A + b \cos B + c \cos C] = a + b + c$
 $= [\sin 2A + \sin 2B + \sin 2C] = \sin A + \sin B + \sin C$ (using $a = 2R \sin A$ etc.)
 $\Rightarrow 4 \sin A \sin B \sin C = \sin A + \sin B + \sin C$
 $\therefore \frac{4abc}{8R^3} = \frac{a+b+c}{2R}$
 $\Rightarrow \frac{abc}{R^2} = (a+b+c) = 2s$
 $\frac{4\Delta}{R} = 2s \left[\frac{abc}{4\Delta} = R \right] \Rightarrow \frac{2\Delta}{s} = R \Rightarrow 2r = R \Rightarrow [B]$
 Also $2r = R \Rightarrow$ triangle is equilateral $\Rightarrow [C]$

Q31. Internal bisector of $\angle A$ of a triangle ABC meets side BC at D. A line drawn through D perpendicular to AD intersects the side AC at E and the side AB at F. If a, b, c represent sides of $\triangle ABC$ then

- (A) AE is HM of b and c (B) $AD = \frac{2bc}{b+c} \cos \frac{A}{2}$ (C) $EF = \frac{4bc}{b+c} \sin \frac{A}{2}$
 (D) the triangle AEF is isosceles

Ans. D, C, B, A

Sol.



(B) $AD = \frac{2bc}{b+c} \cos \frac{A}{2}$ (length of bisector)

(A) in $\triangle ADE$

$$\cos \frac{A}{2} = \frac{AD}{AE}$$

$$AE = \frac{AD}{\cos \frac{A}{2}} = \frac{2bc}{b+c} \text{ (HM of b and c)}$$

(C) $DE = DF$

In $\triangle ADE$

$$\sin \frac{A}{2} = \frac{DE}{AE}$$

$$DE = AE \sin \frac{A}{2} = \frac{A}{2}$$

$$EF = 2AE \sin \frac{A}{2}$$

$$= \frac{4bc}{b+c} \sin \frac{A}{2}$$

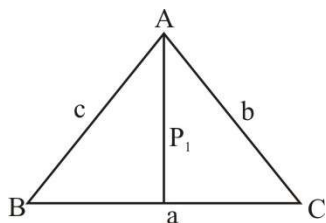
(D) $AE = AF$ so triangle AEF isosceles.

Q32. In $\triangle ABC$ if p_1, p_2, p_3 are lengths of altitudes from vertices A, B, C respectively and r_1, r_2, r_3 denote the radii of ex-circles of $\triangle ABC$, if $a > b > c$, then

- (A) $p_1 > p_2 > p_3$ (B) $p_1 < p_2 < p_3$ (C) $r_1 > r_2 > r_3$ (D) $r_1 < r_2 < r_3$

Ans. B, C

Sol.



$$\Delta = \frac{1}{2} a P_1 \quad P_2 \Delta = \frac{2\Delta}{b}$$

$$P_1 = \frac{2\Delta}{a} \quad P_3 \Delta = \frac{2\Delta}{c}$$

$$\therefore a > b > c$$

$$P_1 < P_2 < P_3$$

$$r_1 = \frac{\Delta}{s-a}, r_2 = \frac{\Delta}{s-b}, r_3 = \frac{2\Delta}{s-c}$$

$$a > b > c$$

$$r_1 > r_2 > r_3$$

Paragraph

In a triangle ABC, let $\tan A = 1$, $\tan B = 2$, $\tan C = 3$ and $c = 3$.

\n

Q33. The radius of the circle circumscribing the triangle ABC, is equal to

- (A) $\frac{\sqrt{10}}{2}$ (B) $\sqrt{5}$ (C) $\sqrt{10}$ (D) $\frac{\sqrt{5}}{2}$

Ans. A

Sol. $\tan A = 1$, $\angle A = 45^\circ$

$$\tan B = 2, \quad \sin B = \frac{2}{\sqrt{5}}$$

$$\tan C = 3, \quad \sin C = \frac{3}{\sqrt{10}}$$

$$C = 3$$

$$\frac{a}{\sin 45} = \frac{\frac{b}{2}}{\sqrt{5}} = \frac{\frac{c}{3}}{\sqrt{10}} = 2R$$

$$R = \frac{\sqrt{10}}{2}$$

$$a = 2R \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{10}}{\sqrt{2}} = 2\sqrt{2}$$

$$R = \frac{3}{\sqrt{10}}$$

Paragraph

In a triangle ABC, let $\tan A = 1$, $\tan B = 2$, $\tan C = 3$ and $c = 3$.

\n

Q34. Let Δ denote the area of the triangle ABC and Δ_p be the area of its pedal triangle.

If $\Delta = k \Delta_p$ then k is equal to

(A) $\sqrt{10}$ (B) $2\sqrt{5}$ (C) 5 (D) $2\sqrt{10}$

Ans. C

Sol. $\tan A = 1$, $\angle A = 45^\circ$

$$\tan B = 2, \quad \sin B = \frac{2}{\sqrt{5}}$$

$$\tan C = 3, \quad \sin C = \frac{3}{\sqrt{10}}$$

$$C = 3$$

$$\frac{a}{\sin 45} = \frac{\frac{b}{2}}{\sqrt{5}} = \frac{\frac{c}{3}}{\sqrt{10}} = 2R$$

$$R = \frac{\sqrt{10}}{2}$$

$$a = 2R \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{10}}{\sqrt{2}} = 2\sqrt{2}$$

$$\begin{aligned} k = \frac{\Delta}{\Delta_p} &= \frac{\frac{1}{2}bc\sin A}{\frac{1}{2}b^1c^1\sin A^1} = \frac{bc\sin A}{\cos B \cdot C\cos C \cdot \sin(\pi \cdot 2A)} \\ &= \frac{\sin A}{\cos B \cdot \cos C \cdot 2\sin A \cos B} \\ &= \frac{1}{\frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{10}} \cdot \frac{1}{\sqrt{2}}} = 5 \end{aligned}$$

Paragraph

In a triangle ABC, let $\tan A = 1$, $\tan B = 2$, $\tan C = 3$ and $c = 3$.

\n

Q35. Area of the triangle ABC is equal to

- (A) $\frac{3\sqrt{2}}{2}$ (B) 3 (C) $2\sqrt{3}$ (D) $3\sqrt{2}$

Ans. B

Sol. $\tan A = 1$, $\angle A = 45^\circ$

$$\tan B = 2, \quad \sin B = \frac{2}{\sqrt{5}}$$

$$\tan C = 3, \quad \sin C = \frac{3}{\sqrt{10}}$$

$$C = 3$$

$$\frac{a}{\sin 45} = \frac{\frac{b}{2}}{\frac{1}{\sqrt{5}}} = \frac{\frac{c}{3}}{\frac{1}{\sqrt{10}}} = 2R$$

$$R = \frac{\sqrt{10}}{2}$$

$$a = 2R \cdot \frac{1}{\sqrt{2}} = \frac{\sqrt{10}}{\sqrt{2}} = 2\sqrt{2}$$

$$\Delta = \frac{1}{2} ab \sin C$$

$$\Delta = \frac{1}{2} \sqrt{5} \cdot 2\sqrt{2} \cdot \frac{3}{\sqrt{10}} = 3$$

Q36. In a ΔABC , $BC = 2$, $CA = 1 + \sqrt{3}$ and $\angle C = 60^\circ$. Feet of the perpendicular from A, B and C on the opposite sides BC, CA and AB are D, E and F respectively and are concurrent at P. Now match the entries of column-I with respective entries of column-II.

Column-I

(A) Radius of the circle circumscribing the ΔDEF , is

(B) Area of the ΔDEF , is

(C) Radius of the circle inscribed in the ΔDEF , is

Column-II

(P) $\frac{\sqrt{6} - \sqrt{2}}{4}$

(Q) $\frac{1}{\sqrt{2}}$

(R) $\frac{\sqrt{3}}{4}$

(S) $\frac{\sqrt{6} + \sqrt{2}}{4}$

(A) (A) Q (B) R (C) P (B) (A) S (B) R (C) P (C) (A) P (B) S (C) Q

(D) (A) Q (B) S (C) R

Ans. 5ef85d5f239ecc768b1bddd8

Sol.

$$\cos 60 = \frac{(2)^2 + (1 + \sqrt{3})^2 - C^2}{2(2)(1 + \sqrt{3})} = \frac{1}{2}$$

$$4 + 1 + 3 + 2\sqrt{3} - C^2 = 2 + 2\sqrt{3}$$

$$C = \sqrt{6}$$

$$\frac{C}{\sin C} = 2R$$

$$\frac{\sqrt{6}}{\frac{\sqrt{3}}{2}} = 2R$$

$$R = \sqrt{2}$$

$$R_P = \frac{R}{2} = \frac{1}{\sqrt{2}}$$

$$\frac{2}{\sin A} = \frac{1 + \sqrt{3}}{\sin B} = 2\sqrt{2}$$

$$\sin A = \frac{1}{\sqrt{2}}, \quad \sin B = \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

$$\angle A = 48^\circ \quad \angle B = 75^\circ$$

$$\Delta DEF = a^1 b^1 \sin(c^1)$$

$$= \frac{1}{2} R \sin 2A \cdot R \sin 2B \cdot \sin(\pi - 2C)$$

$$= \frac{1}{2} (\sqrt{2})^2 \cdot \sin 90^\circ \cdot \sin 150^\circ \cdot \sin 60^\circ$$

$$= (1) \frac{1}{2} \cdot \frac{\sqrt{3}}{2} =$$

Q37. Let P be an interior point of ΔABC .

Match the correct entries for the ratios of the Area of ΔPBC : Area of ΔPCA : Area of ΔPAB

depending on the position of the point P w.r.t. ΔABC .

Column-I

Column-II

(A) If P is centroid (G)

(P) $\tan A : \tan B : \tan C$

(B) If P is incentre (I)

(Q) $\sin 2A : \sin 2B : \sin 2C$

(C) If P is orthocentre (H)

(R) $\sin A : \sin B : \sin C$

(D) If P is circumcentre (S)

(S) $1 : 1 : 1$

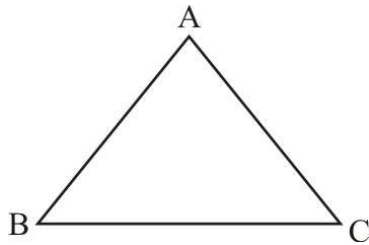
(T) $\cos A : \cos B : \cos C$

(A) (A) R (B) S (C) P (D) Q (B) (A) S (B) R (C) P (D) Q

(C) (A) S (B) R (C) Q (D) P (D) (A) Q (B) P (C) R (D) S

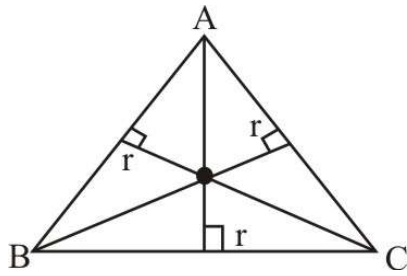
Ans. 5ef85f354c298c76832e6626

Sol. (A) If P is centroid (~)



$$\Delta PBC : \Delta PCA : \Delta PAB = 1 : 1 : 1$$

(B) P is incentre (I)

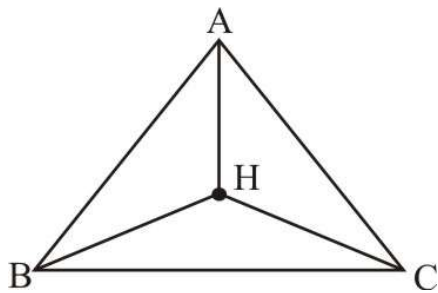


$$\Delta PBC : \Delta PCA : \Delta PAB$$

$$= \frac{1}{2}ar : \frac{1}{2}br : \frac{1}{2}cr$$

$$= \sin A : \sin B : \sin C$$

(C) P is orthocentre



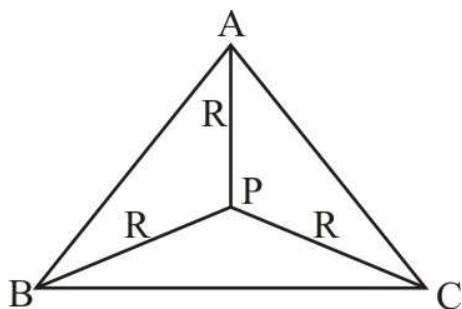
$$\Delta PBC : \Delta PAC : \Delta PAB$$

$$= \frac{1}{2}ar \cdot 2R \cos B \cdot \cos C : \frac{1}{2}b \cdot 2R \cos A \cos C : \frac{1}{2}c \cdot 2R \cos A \cos B$$

$$\text{by } \cos A : \cos B : \cos C$$

$$= \tan A : \tan B : \tan C$$

(D) If P is circumcentre



$$\Delta PBC : \Delta PAC : \Delta PAB$$

$$= \frac{1}{2} R^2 \sin A : \frac{1}{2} R^2 \sin 2B : \frac{1}{2} R^2 \sin^2 C$$

$$= \sin A : \sin B : \sin C$$